



Giovanni August Immanuel Wijaya

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🔗 github.com/ghyoco 🔗 portfolio

Profile

I am a fifth year computer science undergraduate at Bina Nusantara University with practical experience developing computer vision pipelines and predictive machine learning models. Seeking an AI Internship or Software Engineering to apply strong problem-solving skills and pipeline development to real world software solutions.

Education

SMAK Penabur Gading Serpong 07/2021 – 05/2024

Bina Nusantara University, Computer Science 09/2024 – present
GPA: 3.81

Projects

PathFinder 📄, *Real-Time Lane Detection Using Computer Vision* 02/2026 – 06/2026

- Engineered a computer vision pipeline capable of detecting lane boundaries and calculating vehicle lane drift across lightings (day and night).
- Implemented Bird's-Eye View (BEV) inverse perspective mapping, sliding-window clustering, and 2nd-degree polynomial curve fitting ($x = ay^2 + by + c$).
- Designed a FastAPI backend and deployed a live interactive web app with Docker on Hugging Face Spaces.

Smart Building Time-Series Energy Forecasting 📄, 02/2026 – 06/2026

Comparing LSTM vs XGBoost for energy load forecasting on smart buildings

- Built a forecasting pipeline on the CU-BEMS dataset to predict 30-minute building power loads from multi-floor energy and indoor air quality sensor data.
- Built lag, rolling window, and cyclical time features to benchmark LSTM networks against XGBoost using chronological TimeSeriesSplit cross-validation.
- Achieved an R^2 of 0.9818, SMAPE of 6.95%, and 68.1% RMSE reduction over the baseline with XGBoost, validated via Diebold-Mariano testing.

Volunteering Experience

Bimbingan belajar Sitanala, Volunteer 02/2026 – 04/2026
Taught basic math and English lesson to elementary students in Dutasia Sitanala City

Certificates

Pelatihan Azure AI Fundamentals: Microsoft AI-900T00-A Belajar AI dari Dasar

Technical Skills

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|----------------|--------------|----------------|--------------|
| • Python | • Numpy | • pandas | • Docker |
| • OpenCV | • FastAPI | • scikit-learn | • TensorFlow |
| • Git / GitHub | • Matplotlib | • TypeScript | |

Languages

Indonesian — Native/Bilingual	English — Proficient
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